

THE PLAYBOOK OF A “SHARP” POLYMARKET TRADER

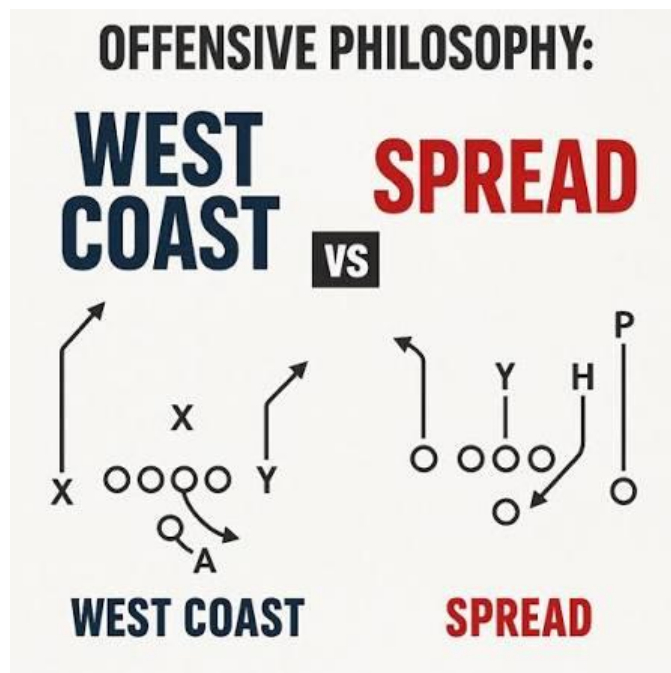
A Behavioral Analysis of 604,578 Polymarket Trader Profiles

Lakshay Naresh | lakshayn@terpmail.umd.edu

Introduction

In football, different teams choose different schemes and playbooks to create an advantage. One team may run a West Coast offense built around short, precise passes and yards after the catch. Another may use a spread offense to create space, stretch the defense, and generate one-on-one matchups. A third may rely on a run-heavy system, aggressive play-action, or a defense-first approach that controls field position and limits risk. These teams are not all trying to win in the same way, and they do not reduce their strategy to one formation or one statistic.

The important point is that a scheme is not successful simply because it is recognizable. Different playbooks can lead to success or failure depending on how well their components fit together, how effectively they are executed, and whether they create an advantage against the competition. A team's pace, player usage, risk tolerance, and allocation of resources all work together to define its style.



The same idea applies to prediction-market trading. Polymarket traders do not all follow the same strategy or display the same tendencies. Some may make many small trades across a wide range of markets, while others may concentrate capital in a few topics and place fewer, larger-conviction trades. Some may specialize in one domain, while others diversify across many categories. Some may trade frequently and aggressively, while others wait for a small number of opportunities.

These differences can be understood as distinct trading schemes or playbooks. Just as different football schemes can produce either success or failure, different trading styles may be associated with stronger or weaker performance. Looking only at total P&L would miss this distinction because trading scale varies dramatically across participants. The more useful question is not simply who traded the most or earned the most dollars, but what behavioral characteristics define different trader playbooks and which of those playbooks are associated with stronger profit per dollar traded.

Data Overview

This project analyzes 604,578 unique Polymarket traders. Each observation is a trader-level profile containing P&L, total volume, transaction activity, market breadth, transaction size, execution-related measures, timing features, and a complete allocation across 17 market topics. The dataset also assigns each trader to one of four labels: “awful,” “bad,” “good,” or “sharp”.

The initial data audit shows a highly heterogeneous market. Median trader volume is approximately \$148, while the top 1% of traders exceed roughly \$113,000 in volume. Total P&L is similarly heavy-tailed, ranging from approximately negative \$1.22 million to positive \$1.10 million. These large differences make raw averages misleading; the analysis will therefore emphasize normalized performance, medians, percentiles, and log-scaled visualizations where appropriate.

Label	Number of traders	Percent
bad	198535	32.8%
good	180422	29.8%
awful	125849	20.8%
sharp	99772	16.5%

Metric	Value
Total trader profiles	604,578
Unique traders	604,578
Variables	41
Exact duplicate rows	0
Number of trader labels	4
Topic-allocation columns	17
Topic shares sum to	100%
Median trading volume	\$148.35
99th percentile trading volume	\$113,129.16
Minimum trader P&L	\$-1,219,870
Maximum trader P&L	\$1,103,256

Research Question

- ◇ Do sharp Polymarket traders converge on one dominant trading playbook, or are multiple distinct behavioral schemes associated with an edge?

More specifically:

- ◇ After accounting for differences in trading scale and activity, how are topic specialization, market breadth, transaction sizing, and execution behavior associated with profit per dollar traded?

Measuring What is *Success*

The primary outcome is “trader_ppv”, which is equal to trader P&L divided by trading volume:

Profit per dollar traded = Trader P&L / Trader Volume

This measure allows traders of very different sizes to be compared on a common basis. Total P&L remains useful for context, but it should not be treated as the main success metric because a large trader can earn more dollars with a weaker rate of return.

The supplied “trader_label” will be used as a secondary validation measure. In particular, I will compare the share of traders labeled “sharp” across behavioral groups. Because the dataset does not document how these labels were constructed, I will not assume that the label is an independent measure of skill. Any relationship between behavior, performance, and labels will be described as an association rather than proof of causation.

Defining a Trader’s Playbook

I define a trader’s playbook as the repeatable pattern in how they deploy capital across Polymarket. Rather than treating every trader as identical, I group their observable behavior into five dimensions: trading tempo, market breadth, conviction and sizing, execution style, and topic specialization.

Together, these dimensions describe how a trader participates in the market. For example, two traders may have similar trading volume but follow very different playbooks: one may make many small trades

across many markets, while another may make a few large, concentrated trades within one topic. The purpose of this framework is to identify whether certain combinations of behavior are associated with stronger normalized profitability.

Playbook dimension	Trading interpretation	Measures used
Tempo	How actively a trader participates	Transactions per day, volume per day, total transaction count
Market breadth	Whether activity is concentrated in a few markets or spread broadly	Markets per day
Conviction and sizing	Whether a trader uses many small positions or fewer larger positions	Mean transaction value, transaction-value variability
Execution style	How aggressively a trader interacts with available market liquidity	Price levels consumed, price levels per transaction, mean delta
Topic specialization	Whether a trader focuses on a narrow domain or diversifies across topics	Topic concentration, entropy, effective number of topics, dominant topic

The 17 topic-allocation variables sum to 100% for every trader. This allows topic specialization to be measured objectively rather than defined subjectively. A trader who allocates nearly all activity to politics, for example, is a concentrated specialist; a trader whose activity is distributed across multiple topics is more diversified.

Methodology

The analysis followed a two-stage design. The first stage examined individual relationships between trading behavior and performance. The second used unsupervised machine learning to identify complete behavioral playbooks. Separating these stages made it possible to answer both parts of the research question: which individual behaviors are associated with performance, and whether traders naturally organize into several broader strategic styles.

Data Preparation and Feature Engineering

The dataset contains 604,578 observations, with each observation representing one unique trader. The audit found no duplicate rows. Missing values occurred primarily in measures of within-trader variability, such as the standard deviations of transaction value, timing, and price movement. Because the main clustering variables were complete for all traders, these missing standard-deviation measures were not imputed or used to determine cluster membership.

Several variables were highly right-skewed. For example, median total trading volume was approximately \$148, while the 99th percentile exceeded \$113,000. A small number of extremely active traders could therefore dominate calculations based on raw values. To reduce this influence, the clustering features were converted into percentile ranks. A trader at the 80th percentile for tempo, for example, was more active than approximately 80% of the traders in the dataset. This transformation preserved the ordering of traders while preventing a small number of extreme observations from determining the cluster structure.

Additional measures were constructed from the 17 topic-allocation columns. These columns sum to 100% for every trader and describe how each trader distributed activity across market categories. Topic specialization was measured using:

- **Top-topic share:** the proportion allocated to the trader’s largest topic.
- **Topic entropy:** the extent to which activity was distributed across several topics.
- **Effective number of topics:** an interpretable measure of topic breadth derived from the concentration of the topic shares.
- **Dominant topic:** the market category receiving the trader’s largest allocation.

A trader with a top-topic share of 100% and an effective-topic count of one was treated as a pure specialist. A trader with activity distributed more evenly across several topics had higher entropy and a larger effective number of topics.

Direct Behavioral Comparisons

Before constructing complete playbooks, I examined several individual behavioral relationships. Traders were grouped according to topic specialization, transactions per day, and execution depth. Within each group, I compared median profit per dollar traded and the percentage of traders labeled “sharp”.

These comparisons provided the conventional independent-variable-versus-dependent-variable component of the analysis:

- **Independent behavioral variables:** topic specialization, trading activity, market breadth, transaction sizing, and execution behavior.
- **Primary dependent variable:** profit per dollar traded, measured by “trader_ppv”.
- **Secondary outcome:** whether the trader was labeled “sharp”.

A logistic regression was also used to estimate the association between specialization and the probability of receiving a “sharp” label while controlling for differences in trading scale and activity. The controls included total trading volume, transaction count, transactions per day, and markets per day. Highly skewed control variables were log-transformed before estimation.

These models were interpreted as measures of association rather than causation. The data show how behaviors and outcomes occur together, but they do not establish that changing a behavior would necessarily cause performance to improve.

Constructing the Playbook Dimensions

The clustering analysis summarized trader behavior using five composite dimensions. Each dimension was constructed from variables representing a related part of trading behavior:

Playbook dimension	Variables represented
Tempo	Transactions per day, volume per day, and total transaction count
Market breadth	Markets traded per day
Conviction sizing	Mean transaction value
Price and execution intensity	Price levels consumed per transaction and mean delta
Topic diversification	Topic entropy and the effective breadth of topic exposure

Where a dimension contained multiple variables, their percentile ranks were averaged to create a single composite score. The resulting scores were standardized before clustering so that no dimension received greater weight merely because it had a larger numerical scale.

Importantly, total P&L, “trader_ppv”, profitability, and “trader_label” were excluded from every clustering input. The algorithm therefore grouped traders exclusively according to behavior. Performance information was introduced only after the clusters had been formed. This separation prevents the model from defining a successful playbook simply because it was given the outcomes in advance.

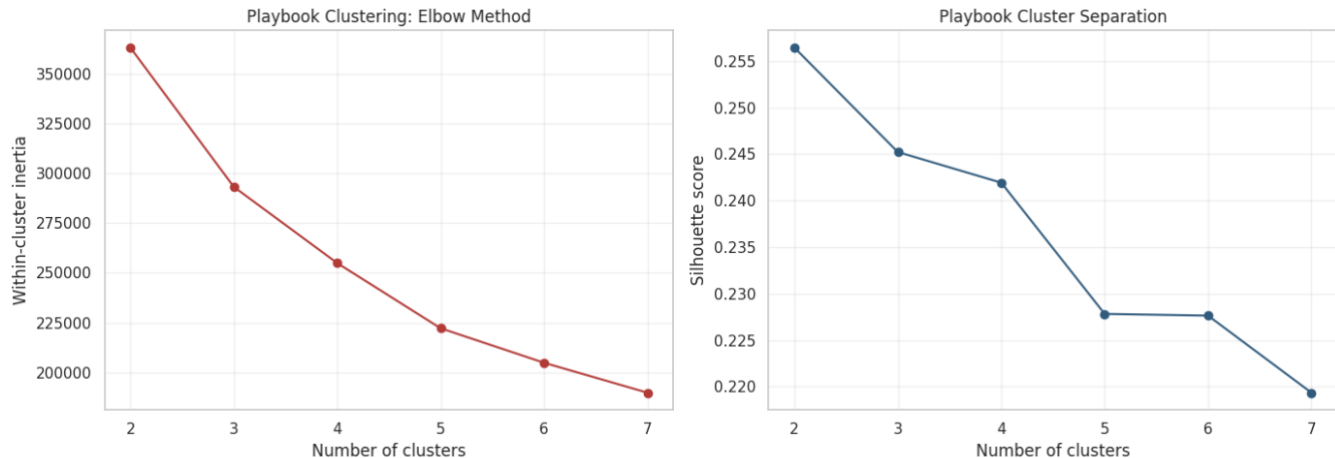
Identifying the Behavioral Playbooks

K-means clustering was used to identify traders with similar combinations of the five behavioral dimensions. K-means assigns traders to clusters by minimizing the within-cluster distance between traders and their cluster center. In practical terms, traders placed in the same cluster displayed similar combinations of tempo, market breadth, position sizing, execution intensity, and topic diversification.

Candidate solutions containing between two and seven clusters were evaluated. Two diagnostic measures were considered:

1. **Within-cluster inertia**, which measures how tightly traders are grouped around their assigned cluster centers. Lower inertia indicates more compact clusters, although it will mechanically decline as more clusters are added.
2. **Silhouette score**, which compares similarity within a cluster with separation from other clusters. Higher scores indicate clearer separation.

Cluster size and interpretability were also considered. A statistically separated solution would have limited practical value if it produced extremely small groups or playbooks that could not be meaningfully distinguished.



The two-cluster solution produced the highest silhouette score, but it divided the market into only two broad styles and concealed important behavioral differences. The three- and four-cluster solutions had similar silhouette scores, approximately 0.245 and 0.242, respectively. Four clusters were selected because the additional cluster produced a more detailed and interpretable set of playbooks without a substantial loss of separation. The final cluster shares were also reasonably balanced, ranging from approximately 21% to 33% of traders.

After selecting four clusters, the model was applied to the complete dataset. Each cluster was then profiled using both its composite scores and the underlying raw variables. Descriptive names were assigned only after examining these behavioral profiles:

- **Low-Tempo Focused**

- **High-Tempo Generalists**
- **Concentrated Conviction**
- **Selective Specialists**

These names are interpretive summaries rather than inputs supplied to the clustering algorithm.

Evaluating Playbook Performance

Once the four behavioral playbooks had been established, their outcomes were compared using measures that were not involved in cluster formation. The primary comparison was median “trader_ppv”, expressed in basis points:

$$\text{PPV in basis points} = \frac{\text{Trader P\&L}}{\text{Trader Volume}} \times 10,000$$

One basis point equals 0.01%. Medians were emphasized because the performance distributions contained large positive and negative outliers. Each playbook was evaluated using:

- Median profit per dollar traded
- Median total P&L and trading volume
- Percentage of traders with positive P&L
- Percentage labeled “sharp”
- Combined percentage labeled “good” or “sharp”
- Distribution percentiles showing downside risk and upside potential

Statistical tests were used to determine whether the observed differences were larger than would be expected from sampling variation alone. A Kruskal–Wallis test compared the “trader_ppv” distributions across the four playbooks because the outcome was highly skewed and unsuitable for a standard comparison of means. Epsilon squared was reported to measure the practical size of the relationship.

A chi-square test examined whether the distribution of the four trader labels differed across playbooks. Cramér’s V was reported as the corresponding effect-size measure. Finally, nonparametric bootstrap confidence intervals were calculated around each playbook’s median profit per dollar traded.

This combination of clustering, outcome comparison, effect sizes, and confidence intervals allowed the analysis to distinguish statistical significance from practical importance. With more than 600,000 observations, even small differences can produce very low p-values. The interpretation therefore focuses not only on whether playbooks differ, but also on how large and economically meaningful those differences appear to be.

Results

The analysis produced two complementary sets of findings. First, direct comparisons showed how individual behavioral characteristics (particularly topic concentration, trading activity, and execution intensity) were associated with performance. Second, the clustering analysis identified four broader trading playbooks that combined these individual characteristics in different ways.

The results do not indicate that all successful traders follow one universally superior strategy. Instead, they reveal meaningful differences between a high-variance playbook that contains many traders labeled “sharp” and a more selective playbook that produces the strongest typical profit per dollar traded.

Individual Behavioral Relationships

Topic Specialization:

Topic concentration did not have a simple linear relationship with performance. Pure specialists, whose activity was entirely allocated to one topic, had a median profit per dollar traded of approximately zero and a “sharp” rate of 14.97%. Traders in the highly concentrated group had the highest unadjusted “sharp” rate at 18.28%, while moderately concentrated and diversified traders recorded rates of 17.01% and 16.81%, respectively.

The very diversified group represented only 754 traders, or approximately 0.12% of the dataset. Its small size means that its results should be interpreted cautiously.

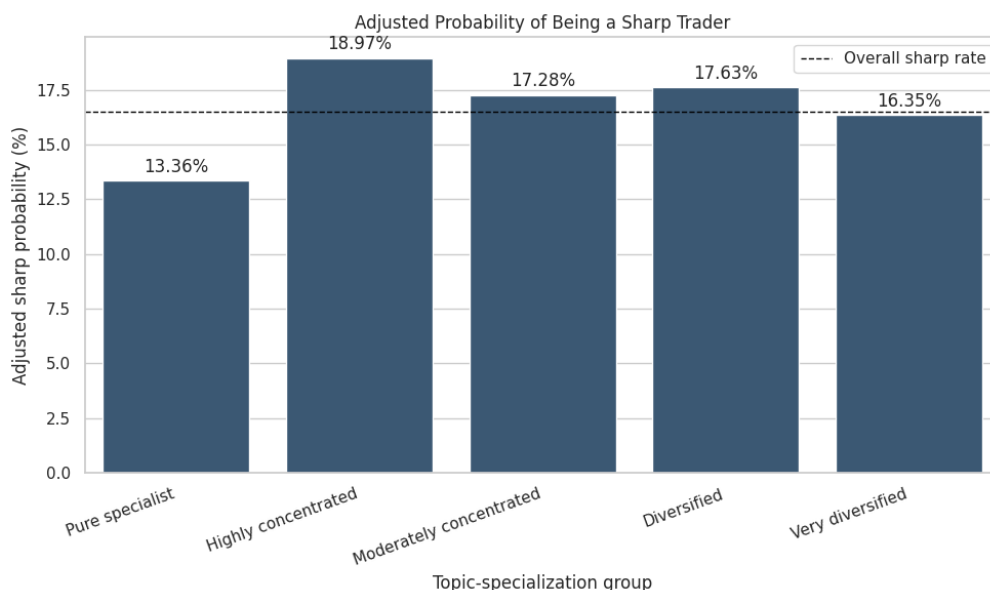
Specialization group	Traders	Median effective topics	Median PPV	Sharp rate
Pure specialist	195,313	1.000	0.000%	14.97%
Highly concentrated	92,251	1.306	−0.081%	18.28%
Moderately concentrated	198,510	2.000	−0.088%	17.01%
Diversified	117,750	3.220	−0.120%	16.81%
Very diversified	754	5.432	−0.072%	15.65%

The raw comparison therefore provides only limited support for the specialization hypothesis. Complete specialization did not produce the highest “sharp” rate, and greater diversification did not consistently improve median profit per dollar traded.

The adjusted model produced a more nuanced result. After controlling for volume, transaction count, transactions per day, and markets per day, highly concentrated traders had 1.52 times the odds of being labeled “sharp” compared with pure specialists. Moderately concentrated and diversified traders had odds ratios of 1.36 and 1.39, respectively.

The adjusted probability of receiving a “sharp” label was 13.36% for pure specialists and 18.97% for highly concentrated traders. The corresponding probabilities were 17.28% for moderately concentrated traders, 17.63% for diversified traders, and 16.35% for very diversified traders.

These results suggest that some degree of breadth may be advantageous, but unrestricted diversification is not clearly superior. The highest adjusted probability occurred among traders who were strongly concentrated without being restricted entirely to one topic.



Trading Activity:

Trading more frequently was not associated with a steadily increasing edge. Traders in the bottom 25% of transactions per day completed a median of one transaction per day and earned a median of 29.66 basis points per dollar traded. Their “sharp” rate was 21.70%.

As activity increased, both measures generally weakened. The median profit per dollar traded fell below zero for the middle activity groups, reaching −26.80 basis points among traders in the 75th–90th percentile of activity. The percentage labeled “sharp” declined almost monotonically from 21.70% in the bottom activity quartile to only 2.76% among the top 1% most active traders.

Activity group	Median transactions per day	Median PPV	Sharp rate
Bottom 25%	1.00	29.66 bps	21.70%
25th–50th percentile	1.67	−3.69 bps	18.64%
50th–75th percentile	2.67	−20.02 bps	14.22%
75th–90th percentile	4.27	−26.80 bps	11.40%
90th–99th percentile	10.00	−21.87 bps	8.31%
Top 1%	58.00	−0.02 bps	2.76%

The top 1% recovered to approximately break-even median performance despite having the lowest “sharp” rate. This indicates that activity, normalized profitability, and the supplied label are capturing different aspects of trader behavior. Overall, however, the results provide little evidence that a higher trading frequency alone produces stronger performance.



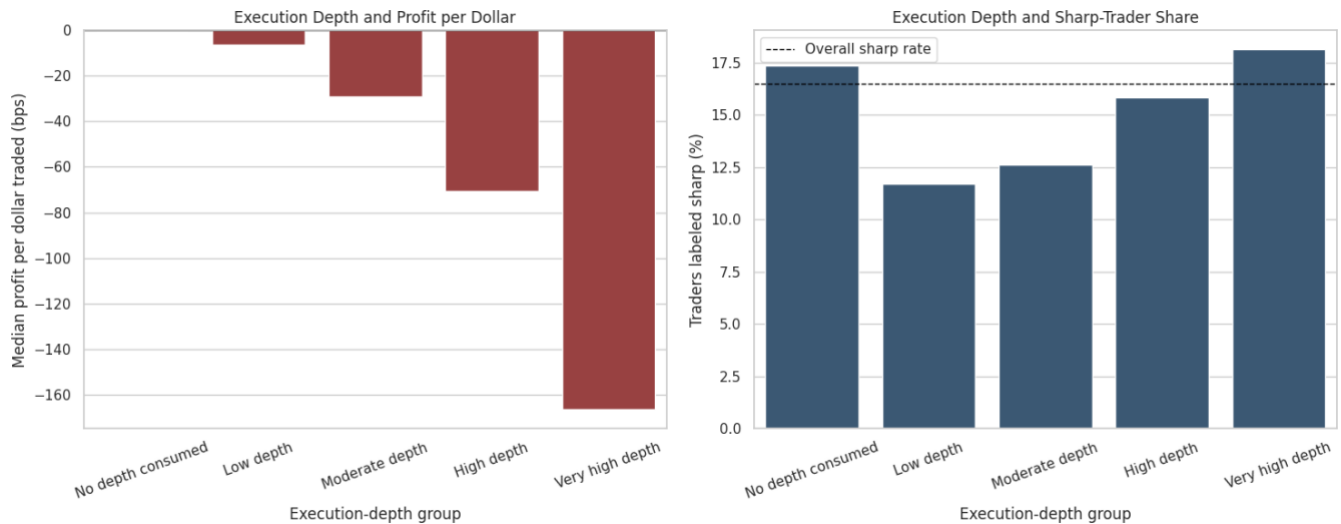
Execution Intensity:

Execution behavior also produced a nonlinear result. Traders recorded as consuming no observable price depth had approximately break-even median profit per dollar traded and a “sharp” rate of 17.36%. Among traders with positive depth consumption, median performance deteriorated as execution depth increased.

The low-depth group had a median PPV of −6.13 basis points. This declined to −29.23 basis points for moderate depth, −70.43 basis points for high depth, and −166.70 basis points for very high depth.

However, the “sharp” rate followed a U-shaped pattern rather than declining continuously. It fell to 11.72% in the low-depth group before increasing to 18.13% among traders with very high execution depth. The very-high-depth group therefore combined the worst typical normalized performance with the highest “sharp” rate among traders consuming positive depth.

This pattern is important because it shows that aggressive execution may characterize both strong informed traders and traders making costly or poorly timed trades. Execution intensity alone is therefore insufficient to distinguish skill. Its meaning depends on how it is combined with sizing, tempo, and specialization within a complete playbook.

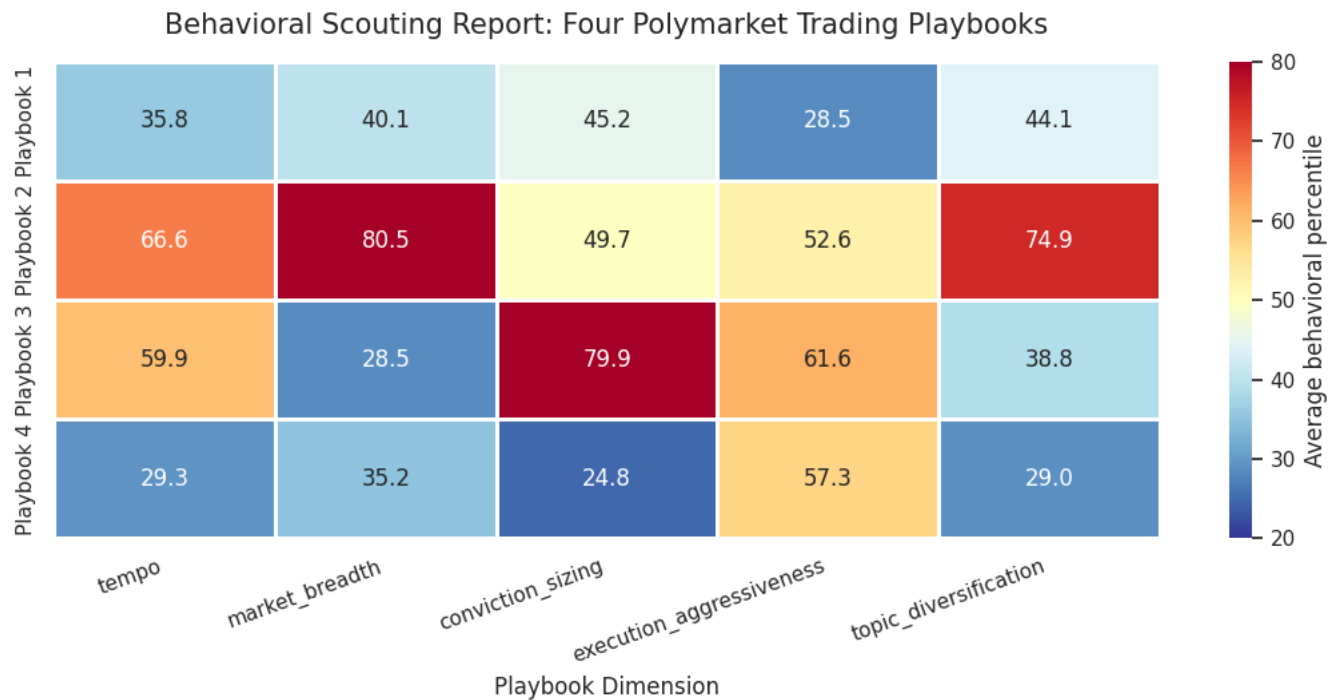


Four Behavioral Playbooks

The four-cluster solution separated the 604,578 traders into reasonably balanced groups. The largest cluster represented 33.42% of traders, while the smallest represented 21.33%. This balance allowed the analysis to compare economically meaningful groups rather than treating a small collection of extreme traders as an entire playbook.

Playbook	Traders	Share of traders
Low-Tempo Focused	140,142	23.18%
High-Tempo Generalists	202,071	33.42%
Concentrated Conviction	133,432	22.07%
Selective Specialists	128,933	21.33%

Each playbook exhibited a different combination of tempo, breadth, sizing, execution intensity, and topic diversification.



The heatmap demonstrates that the clusters were differentiated by combinations of characteristics rather than by a single variable.

Playbook 1: Low-Tempo Focused:

Low-Tempo Focused traders represented 23.18% of the sample. They traded at a relatively slow pace, with a median of 1.33 transactions and approximately \$18.85 in volume per day. Their median transaction value was \$6.53, and their median effective number of topics was 1.50.

This playbook scored below the overall median in tempo, market breadth, price and execution intensity, and topic diversification. Its members were relatively inactive and focused, but they were not as completely specialized as the Selective Specialists.

Playbook 2: High-Tempo Generalists:

High-Tempo Generalists formed the largest cluster, representing 33.42% of traders. Their median activity was 3.36 transactions and \$58.73 in volume per day. They also participated in a median of two markets per day, compared with one for each of the other playbooks.

This group received the highest market-breadth score, at the 80.5th percentile, and the highest topic-diversification score, at the 74.9th percentile. Its median effective number of topics was 2.56, the largest of the four groups. These traders can therefore be characterized as active generalists who spread their participation across more markets and topics.

Playbook 3: Concentrated Conviction:

Concentrated Conviction traders represented 22.07% of the sample. Their defining characteristic was transaction size. Their conviction-sizing score was at the 79.9th percentile, substantially above every other playbook.

The median transaction value for this group was \$49, compared with less than \$8 for each of the other playbooks. Its median volume per day was approximately \$194, also the highest among the four groups.

At the same time, its median effective-topic count was only 1.11, and its median top-topic share was 94.8%.

This playbook therefore combined relatively large positions with concentrated topic exposure. Its members did not trade across the broadest range of markets, but they deployed substantially more capital when they participated.

Playbook 4: Selective Specialists:

Selective Specialists represented 21.33% of traders. This group had the lowest tempo score, the smallest median transaction value, and the most concentrated topic profile.

The typical member completed one transaction per day, generated \$11.52 in daily volume, and had a mean transaction value of \$2. Its median effective-topic count was exactly one, and its median top-topic share was 100%. This means that at least half of the traders in this playbook allocated all observed topic activity to a single category.

Despite consuming relatively little observable order-book depth, the group had a high price and execution-intensity score because of its comparatively high “mean_delta”. For this reason, the dimension should be understood as broader price and execution intensity rather than order-book depth alone.

The four playbooks can be summarized as follows:

Playbook	Defining behavioral pattern
Low-Tempo Focused	Slow, focused participation with relatively small trades
High-Tempo Generalists	Frequent activity across the broadest set of markets and topics
Concentrated Conviction	Larger positions concentrated in a narrow set of topics
Selective Specialists	Very low activity, very small trades, and almost complete topic specialization

Performance Across Playbooks

The four behavioral playbooks produced substantially different performance profiles.

Playbook	Median P&L	Median volume	Median PPV	Profitable traders	Sharp rate	Good or sharp
Low-Tempo Focused	−\$1.00	\$58.93	−154.11 bps	37.52%	29.22%	38.07%
High-Tempo Generalists	−\$0.40	\$337.04	−14.06 bps	40.42%	14.33%	41.28%
Concentrated Conviction	−\$0.07	\$612.00	−3.50 bps	44.74%	12.02%	46.08%
Selective Specialists	\$0.00	\$24.00	6.05 bps	54.51%	10.73%	63.55%

Selective Specialists produced the strongest typical performance. They were the only playbook with a positive median profit per dollar traded, earning 6.05 basis points. They also had the highest percentage

of profitable traders, at 54.51%, and the highest combined percentage of “good” or “sharp” traders, at 63.55%.

Concentrated Conviction traders produced the second-best typical result. Their median PPV was slightly negative at -3.50 basis points, and 44.74% were profitable. This group also had the highest median trading volume at \$612, meaning its approximately break-even normalized performance was generated at a much larger scale than that of the Selective Specialists.

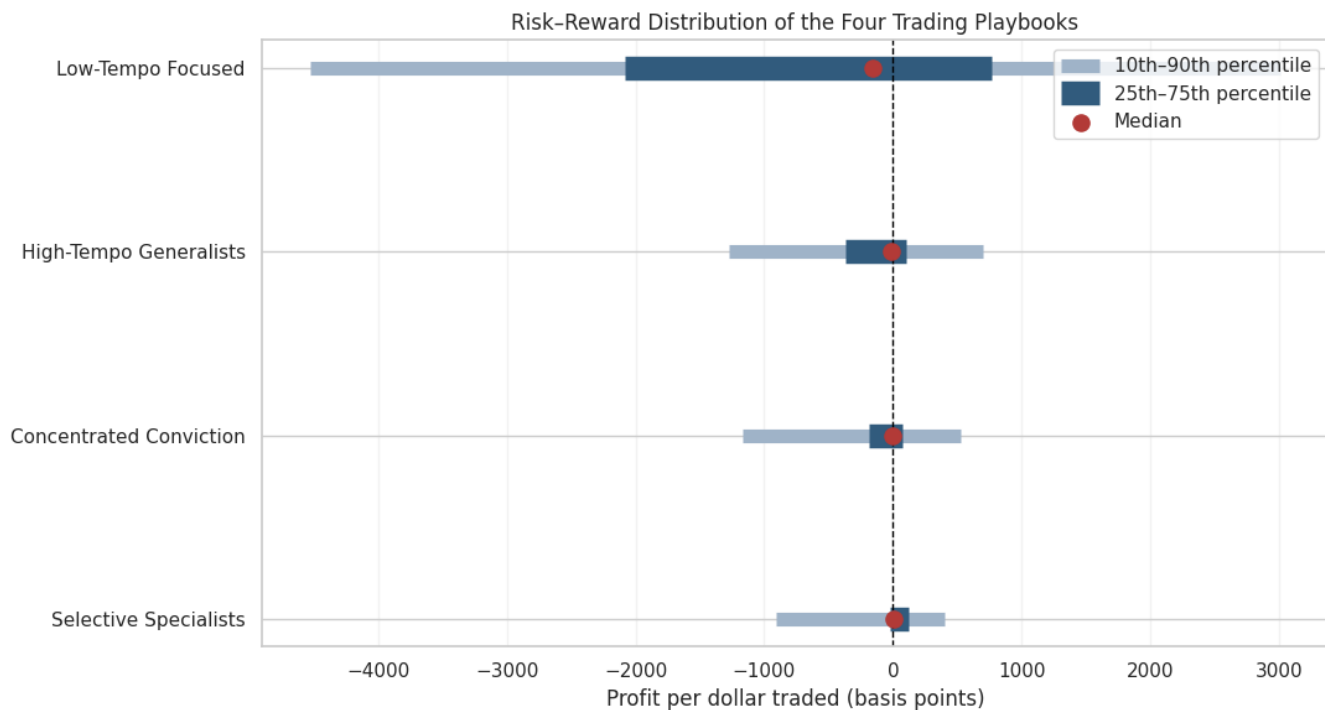
High-Tempo Generalists recorded a median PPV of -14.06 basis points and a profitable-trader share of 40.42%. Their performance was weaker than that of Concentrated Conviction and Selective Specialists, despite their greater market and topic breadth.

Low-Tempo Focused traders had the weakest median result by a substantial margin. Their median PPV was -154.11 basis points, and only 37.52% were profitable. However, this same group had the highest percentage of traders labeled “sharp”, at 29.22%.

This apparent contradiction is one of the most important findings in the analysis. The playbook with the highest “sharp” rate did not have the strongest typical return. Examining the entire performance distribution helps explain why.

Risk and Reward Profiles

Median performance alone does not capture the degree of variation within each playbook. The distribution of profit per dollar traded shows that the Low-Tempo Focused group was substantially more dispersed than the other groups.



Low-Tempo Focused traders had a median PPV of -154.11 basis points, but their distribution extended from $-4,525.61$ basis points at the 10th percentile to $3,006.59$ basis points at the 90th percentile. This enormous range indicates a high-risk, high-reward playbook containing both extreme winners and extreme losers.

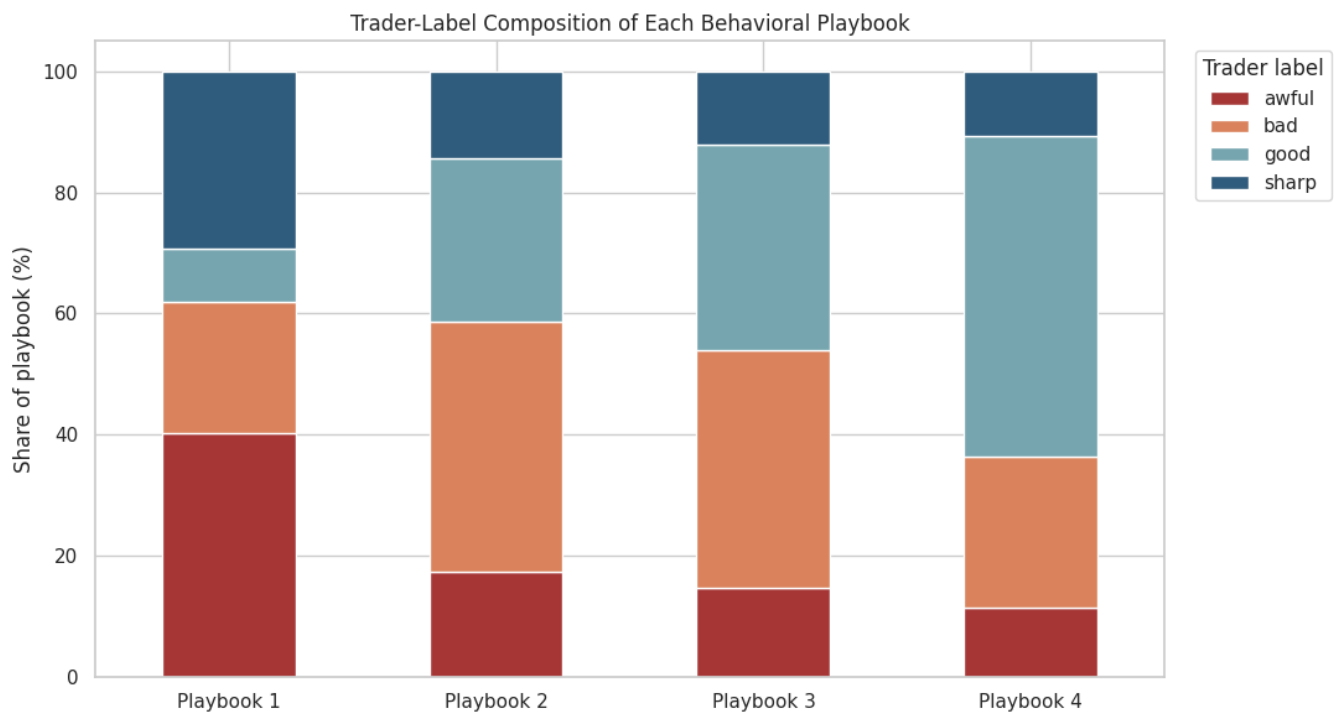
High-Tempo Generalists had a narrower distribution, ranging from −1,271.94 basis points at the 10th percentile to 696.14 basis points at the 90th percentile. Concentrated Conviction traders ranged from −1,172.69 to 520.78 basis points over the same percentiles.

Selective Specialists had the smallest central spread and the only positive median. Their 25th percentile was −20 basis points, their median was 6.05 basis points, and their 75th percentile was 124.10 basis points. Their 90th percentile was lower than that of the other groups, but their downside was also more contained.

The risk–reward figure therefore reveals two distinct forms of apparent success. Low-Tempo Focused traders offered the greatest upside but also the most severe downside. Selective Specialists produced less extreme upside but the strongest and most consistent typical outcome.

Trader-Label Composition

The composition of the supplied trader labels further demonstrates that the playbooks capture different outcome profiles.



Playbook	Awful	Bad	Good	Sharp
Low-Tempo Focused	40.33%	21.60%	8.85%	29.22%
High-Tempo Generalists	17.35%	41.37%	26.95%	14.33%
Concentrated Conviction	14.71%	39.20%	34.07%	12.02%
Selective Specialists	11.36%	25.10%	52.82%	10.73%

Low-Tempo Focused was the most polarized playbook. It contained the largest “sharp” share, at 29.22%, but also the largest “awful” share, at 40.33%. Only 8.85% of the group was labeled “good”. In total, almost 70% of its members occupied one of the two extreme categories.

High-Tempo Generalists were concentrated primarily in the “bad” category, which represented 41.37% of the group. A further 26.95% were labeled “good”, while 14.33% were labeled “sharp”.

Concentrated Conviction traders had a more favorable composition. The “good” category accounted for 34.07% of the playbook, compared with 39.20% labeled “bad”, 14.71% labeled “awful”, and 12.02% labeled “sharp”.

Selective Specialists had the highest “good” share, at 52.82%, and the lowest “awful” share, at 11.36%. Although only 10.73% were labeled “sharp”, the group had the largest combined “good” or “sharp” share.

These label distributions clarify why “sharp” rate should not be treated as the sole measure of playbook quality. Low-Tempo Focused produced a large number of sharp traders, but it also produced the greatest concentration of awful outcomes. Selective Specialists produced fewer extreme winners, but a much larger majority achieved a favorable classification.

Statistical Evidence

The differences in profit per dollar traded across the four playbooks were statistically significant. The Kruskal–Wallis test produced a statistic of $H(3) = 12,425.54$, with $p < 0.001$. This rejects the hypothesis that all four playbooks have identical PPV distributions.

However, the epsilon-squared effect size was 0.021. This is a small effect, indicating that behavioral playbook explains only a limited portion of the total trader-to-trader variation in normalized performance. Playbook membership is associated with performance, but it does not determine an individual trader’s outcome.

The relationship between playbook and trader-label composition was stronger. The chi-square test produced $\chi^2(9) = 112,022.95$, with $p < 0.001$. Cramér’s V was 0.249, indicating a meaningful association between behavioral playbook and the supplied classification.

Because the sample contains more than 600,000 traders, statistical significance alone is not sufficient evidence of practical importance. The effect sizes show that the playbooks are more strongly related to label composition than to the precise level of profit per dollar traded.

Bootstrap confidence intervals provide additional evidence that the playbooks have different typical outcomes:

Playbook	Median PPV	95% confidence interval
Low-Tempo Focused	−154.11 bps	[−162.77, −150.02]
High-Tempo Generalists	−14.06 bps	[−14.87, −13.30]
Concentrated Conviction	−3.50 bps	[−3.83, −3.19]
Selective Specialists	6.05 bps	[5.00, 7.59]

The confidence intervals do not overlap, and the interval for Selective Specialists remains entirely above zero. In this sample, Selective Specialists therefore produced a statistically positive median profit per dollar traded. Nevertheless, this result should be interpreted as an observed association rather than proof that adopting this behavior would cause a trader to become profitable.

Results Summary

The results provide evidence against the idea that all strong traders converge on one universal playbook. Instead, the market contains several behavioral schemes with different risk and performance profiles.

The Low-Tempo Focused playbook generated the greatest polarization: it contained both the largest “sharp” share and the largest “awful” share. The High-Tempo Generalist playbook participated broadly but did not achieve superior normalized performance. Concentrated Conviction traders operated at the greatest scale and approached break-even median efficiency. Selective Specialists produced the strongest typical result, the highest profitable-trader share, and the largest combined percentage of “good” or “sharp” traders.

The central empirical distinction is therefore between **extreme upside** and **consistent efficiency**. The playbook most likely to contain a trader labeled “sharp” was not the playbook with the best median outcome. This distinction forms the basis for the following discussion of what it means to define a successful Polymarket trading scheme.

Discussion/Conclusion

No Single Playbook Defines Success

The central research question asked whether sharp Polymarket traders converge on one dominant behavioral scheme. The results suggest that they do not. Instead, the data reveal several distinct playbooks with different combinations of activity, market breadth, position sizing, execution intensity, specialization, and risk.

This finding closely matches the football analogy motivating the analysis. A team using an aggressive vertical passing offense may generate more explosive plays, but it may also experience greater inconsistency and turnover risk. A team using a conservative ball-control offense may produce fewer spectacular outcomes while succeeding more consistently. Neither scheme can be evaluated from a single statistic because each creates a different distribution of possible outcomes.

The same distinction appears among Polymarket traders. Low-Tempo Focused traders produced the largest concentration of “sharp” traders, but they also produced the largest concentration of “awful” traders and the weakest median profit per dollar traded. Selective Specialists were much less likely to receive the “sharp” label, but they achieved the only positive median PPV, the highest profitable-trader share, and the highest combined percentage of “good” or “sharp” traders.

Success therefore depends on how it is defined. If success means maximizing the probability of an extreme positive classification, the Low-Tempo Focused playbook appears attractive. If success means maximizing the typical trader’s efficiency and probability of finishing profitably, the Selective Specialist playbook appears stronger.

The Difference Between Upside and Consistency

The most important result is the separation between extreme upside and consistent efficiency.

Low-Tempo Focused traders had a median PPV of –154.11 basis points, yet 29.22% were labeled “sharp”. Their PPV distribution was substantially wider than that of the other playbooks, extending from approximately –4,526 basis points at the 10th percentile to more than 3,007 basis points at the 90th

percentile. This is not a uniformly unsuccessful strategy. It is a highly polarized strategy that produces both exceptional winners and severe losers.

Selective Specialists showed the opposite pattern. Their median PPV was positive at 6.05 basis points, 54.51% were profitable, and 63.55% were classified as either “good” or “sharp”. Their upside was less extreme, but their central performance distribution was considerably more stable. Their interquartile range extended from –20 to 124 basis points, compared with approximately –2,080 to 765 basis points for Low-Tempo Focused traders.

In football terms, Low-Tempo Focused resembles an explosive but volatile scheme capable of creating game-changing plays while also producing frequent negative outcomes. Selective Specialists resemble a conservative offense designed to sustain drives, protect possession, and generate smaller but more repeatable gains.

The data do not establish that one approach is universally superior. They show that the two approaches produce different risk–reward profiles. The preferable playbook would depend on a trader’s objectives, risk tolerance, capital constraints, and ability to survive large losses.

Why the Sharp Label and PPV Tell Different Stories

The disagreement between “sharp” rate and median PPV requires careful interpretation. If the supplied label were a complete and independent measure of trading skill, the playbook with the highest “sharp” rate would be expected to perform strongly on normalized profitability. That did not occur.

There are several possible explanations. First, the label may incorporate total profits, trading scale, extreme outcomes, or characteristics not captured by median PPV. A trader can generate a large dollar profit while earning a weaker return per dollar traded, particularly when operating at high volume.

Second, the label may be designed to identify traders with unusually strong upside rather than traders with the most consistent performance. This would explain why the high-variance Low-Tempo Focused playbook contains both the greatest “sharp” share and the greatest “awful” share.

Third, a single trader-level PPV measure cannot distinguish repeatable skill from one unusually successful position. Without transaction-level histories, it is impossible to determine whether an extreme outcome resulted from a persistent edge or a small number of high-impact trades.

For these reasons, “trader_label” is best treated as a secondary validation measure rather than the definitive dependent variable. The strongest assessment of a playbook comes from examining several outcomes together: median PPV, the full PPV distribution, profitable-trader share, total scale, and label composition.

Evaluation of the Hypotheses

The specialization hypothesis received partial support. Complete specialization was not automatically associated with a higher “sharp” rate. In the direct comparison, highly concentrated traders had the highest adjusted probability of receiving a “sharp” label. However, the Selective Specialist playbook produced the strongest median PPV and the largest profitable-trader share. This suggests that specialization may be most useful when combined with selective participation and controlled sizing rather than treated as an isolated characteristic.

The activity-tradeoff hypothesis was supported. Performance did not improve continuously with trading frequency. The least-active quartile had the strongest median PPV in the direct activity comparison,

while the percentage labeled “sharp” declined as transactions per day increased. High activity may allow traders to exploit more opportunities, but activity alone did not appear to create an informational advantage.

The execution hypothesis produced mixed evidence. Greater execution depth was associated with progressively weaker median PPV, which is consistent with aggressive trading incurring greater price impact or occurring when liquidity is limited. At the same time, the “sharp” rate increased again among the most aggressive traders. Aggressive execution may therefore be used by informed traders with strong convictions, but similar behavior may also produce substantial losses when that conviction is incorrect.

The multiple-playbook hypothesis received the strongest support. The clustering analysis identified four behaviorally distinct and reasonably balanced groups. Their performance distributions and label compositions differed significantly, demonstrating that traders labeled “sharp” do not belong to one homogeneous behavioral category.

Implications for Evaluating Traders

The findings suggest that trader evaluation should separate style from skill. Tempo, concentration, transaction size, and execution intensity describe how a trader participates, but none independently proves that the trader possesses an edge.

A useful scouting report should therefore answer two different questions:

1. What playbook does the trader follow?
2. How effectively does the trader execute that playbook?

Two traders can display nearly identical observable behavior and experience very different results. This is especially visible in the Low-Tempo Focused group, where the same broad scheme contains many of the best and worst classifications.

The framework may be more valuable as a diagnostic tool than as a trading rule. It can identify whether a trader behaves like a selective specialist, an active generalist, or a concentrated conviction trader. Performance measures can then determine whether that individual executes the style successfully.

The analysis also highlights the danger of selecting traders based only on total P&L or a supplied classification. A high-dollar winner may have taken substantially more risk, traded at a much larger scale, or benefited from an unusually favorable outcome. Normalized performance and distributional measures provide a more complete view of the efficiency and consistency of that success.

Limitations and Alternative Explanations

The results are based on aggregated trader-level profiles. The dataset does not provide the complete sequence of individual trades, entry and exit prices, market-resolution dates, or the information available to traders when decisions were made. As a result, the analysis cannot directly measure forecasting accuracy, timing skill, liquidity provision, or whether a trader’s performance persisted through time.

The absence of a documented definition for “trader_label” is another important limitation. The observed disagreement between label composition and median PPV could reflect a deliberate feature of the labeling system, but it could also result from the label using information unavailable in this dataset.

The clustering solution is also descriptive rather than permanent. K-means places every trader into the nearest group even when a trader lies near the boundary between two playbooks. The four clusters should therefore be understood as useful behavioral archetypes rather than rigid categories.

Finally, the analysis does not establish causality. Selective specialization is associated with stronger median PPV, but instructing a trader to reduce activity and focus on one topic would not necessarily reproduce the result. Unobserved variables such as experience, research quality, capital, information access, or market timing could influence both the observed behavior and the resulting performance.

A stronger follow-up study would use transaction-level panel data to test whether playbook membership remains stable over time, whether traders switch playbooks after gains or losses, and whether the identified relationships hold on future markets. Out-of-sample validation would also help determine whether the four playbooks represent persistent trading styles rather than patterns specific to this dataset.

Overall, the results support a view of prediction-market performance in which behavior creates a strategic identity, but execution determines whether that identity succeeds. A playbook describes the scheme; it does not guarantee the outcome.